

Final Exam

Question 1:

D:\AA_FinalExam\hypotenuseCalc.m

```

1  %{
2  Final Exam question 1 part A
3  Alexander Avendano
4  12/08/2025
5  write a function to calculate a hypotenuse given two sides
6  %{
7  %give appropriate name, input variables, and output variables
8  function [c] = hypotenuseCalc(a,b)
9      c = sqrt(a^2 + b^2);
10 end

```

```

>> hypotenuseCalc(3,4)
ans =
    5
>>

```

D:\AA_FinalExam\Final1B.m

```

1  %{
2  Final Exam question 1 part B
3  Alexander Avendano
4  12/08/2025
5  write a function to calculate a hypotenuse given two sides
6  %{
7  anonymousHypotenuse = @(a, b) sqrt(a^2 + b^2);

```

```

>> anonymousHypotenuse(3,4)
ans =
    5
>>

```

Question 2:

D:\AA_FinalExam\Final2.m

```

1  %{
2  Final Exam Problem #2
3  Alexander Avendano
4  12/08/2025
5  create a list of 10 random numbers and output the biggest and smallest
6  along with their index
7  %{
8  intList = []
9  for i = 1:10
10     intList = [intList, floor(100*rand)];
11 end
12 smallI = 1; %save smallest index
13 largeI = 1; %save largest index
14 check_num = 2;
15 while(check_num <= length(intList))
16     if(intList(check_num) < intList(smallI)) %check next number against smallest number
17         smallI = check_num;
18     end
19     if(intList(check_num) > intList(largeI)) %check next number against largest number
20         largeI = check_num;
21     end
22     check_num = check_num + 1;
23 end
24 smallest = intList(smallI);
25 largest = intList(largeI);
26 fprintf('Smallest number in list is %d at index %d\n', smallest, smallI);
27 fprintf('Largest number in list is %d at index %d\n', largest, largeI);

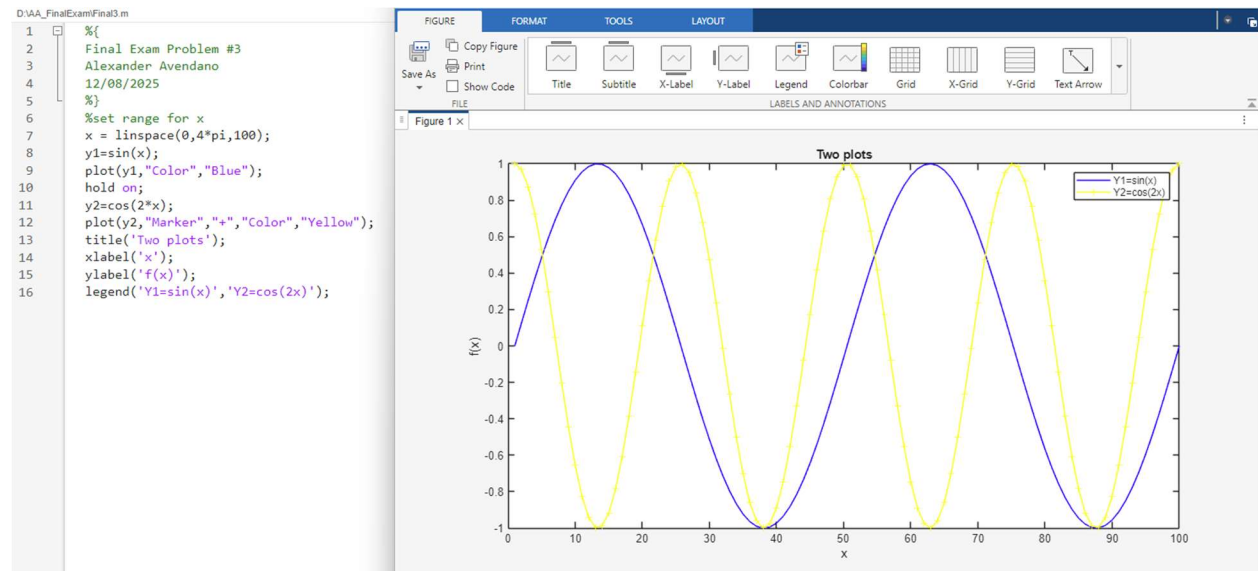
```

```

>> Final2
intList =
[]
- Smallest number in list is 14 at index 6
- Largest number in list is 97 at index 2
>>

```

Question 3:



Question 4:

D:\AA_FinalExam\Final4.m

```

1 %{
2   Final Exam Problem #4
3   Alexander Avendano
4   12/08/2025
5   Create structure array with two classes for next semester
6 %}
7 Course_Schedule(1) = struct('CRN', '12345', 'CourseNumber', ...
8   'ENGL1102', 'Title', 'English 2', 'CreditHours', 3, 'Instructor', ...
9   'Dr. Smith', 'Time', 'TR 8:00-9:15', 'Location', 'Room 101');
10 Course_Schedule(2) = struct('CRN', '54321', 'CourseNumber', ...
11   'Chem3207', 'Title', 'Chem 2', 'CreditHours', 3, 'Instructor', ...
12   'Dr. John', 'Time', 'TR 9:30-10:45', 'Location', 'Room 102');

```

1×2 struct with 7 fields

Fields	CRN	CourseNumber	Title	CreditHours	Instructor	Time	Location
1	'12345'	'ENGL1102'	'English 2'	3	'Dr. Smith'	'TR 8:00-9:15'	'Room 101'
2	'54321'	'Chem3207'	'Chem 2'	3	'Dr. John'	'TR 9:30-10:45'	'Room 102'
3							
4							
5							
6							
7							
8							
9							
10							

Question 5:

D:\AA_FinalExam\Final5A.m

```
1  %{  
2  Final Exam Problem #5A  
3  Alexander Avendano  
4  12/08/2025  
5  %}  
6  %initialize arduino and lcd  
7  a=arduino("COM5","Uno","Libraries","ExampleLCD/LCDAddon")  
8  lcd=addon(a,"ExampleLCD/LCDAddon","RegisterSelectPin","D13","EnablePin", ...  
9  "D12","DataPins",{ "D11","D10","D9","D8"});  
10 initializeLCD(lcd,'Rows',2,'Columns',2);  
11  
12 %print 0 to 9 and 9 to 0  
13 for i=0:9  
14     printLCD(lcd,int2str(i));  
15     printLCD(lcd,int2str(9-i));  
16     clearLCD(lcd);  
17 end
```

D:\AA_FinalExam\Final5B.m

```
1  %{  
2  Final Exam Problem #5B  
3  Alexander Avendano  
4  12/08/2025  
5  %}  
6  %have LED write SOS  
7  on = true;  
8  while (on)  
9      for i=1:3  
10         pause(0.5)  
11         writeDigitalPin(a,"D2",1);  
12         pause(0.5)  
13         writeDigitalPin(a,"D2",0);  
14     end  
15     pause(1.5)  
16     for i=1:3  
17         pause(0.5)  
18         writeDigitalPin(a,"D2",1);  
19         pause(1.5)  
20         writeDigitalPin(a,"D2",0);  
21     end  
22     pause(1.5)  
23     for i=1:3  
24         writeDigitalPin(a,"D2",1);  
25         pause(0.5)  
26         writeDigitalPin(a,"D2",0);  
27         pause  
28     end  
29 end
```